MIL-E-51390A(EA)
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MILITARY SPECIFICATION

ELECTRONIC MODULE ASSEMBLY

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes requirements for one type of electronic module assembly.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

Fed-Std-123 - Marking for Domestic Shipment (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-252 - Wired Equipment, Classification of Visual and Mechanical Defects for.
MIL-STD-810 - Environmental Test Methods.

FSC 6665

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

TDPL 5-15-4550 - Electronic Module Assembly.
D136-15-190 - Tester, Functional, Electronic Module, Q199.

(Copies of specifications, standards, drawings and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 5-15-4550 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings and in this specification.

3.1.2 Components. All components of the electronic module assembly shall conform to the specifications and drawings listed on TDPL 5-15-4550 and subsidiary drawings.

3.2 Manufacture and assembly. The components shall be manufactured and assembled as specified on the subsidiary drawings listed on Drawing D5-15-4550.

3.3 Performance. The performance requirements in this section shall be attained with an input of 24.0 ± 0.5 volt direct current (vdc) unless otherwise specified.

3.3.1 Current drain. The current drain of the electronic module shall not exceed 25.0 milliamperes when tested as specified in 4.4.4.1.

3.3.2 Battery test circuit. The current through the battery test circuit shall be 54.3 2± 3.0 microamperes when tested as specified in 4.4.4.2.

3.3.3 Supply voltage and regulation. The output of the regulated power supply shall be 17.0 ± 0.1 vdc when tested as specified in 4.4.4.3. In addition, the output of the regulated power supply shall not vary more than 0.1 volt as the input voltage is varied from 24.0 ± 0.5 to 38.0 ± 0.5 volt.

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3.3.4 Zero adjust range. The zero adjust control shall compensate for input voltages of at least +48.0 ± 0.5 and -6.0 ± 0.5 millivolt direct current when tested as specified in 4.4.4.4.

3.3.5 Alarm functioning.

3.3.5.1 Actuation.

(a) False alarm. There shall be no more than 1.0 vdc across pins H and Q under each of the following two conditions when tested as specified in 4.4.4.5(a): first when pin H is grounded and second when pin Q is connected to 24 vdc.

(b) Trigger level. With the input of a 15.0 ± 1.0 millivolt trigger, a voltage of 22.0 ± 2.5 vdc shall appear across pins H and Q, and across pins R and T of the electronic module connector when the alarm circuit is actuated as specified in 4.4.4.5(a).

3.3.5.2 Alarm level stability. The millivolt input required to trigger the electronic module in 3.3.5.1 shall not change more than 1.0 millivolt when the electronic module is operated from a supply voltage of 18.5 ± 0.5 volt when tested as specified in 4.4.4.5(b).

3.3.5.3 Latching. The alarm condition established in 3.3.5.1 shall remain as the input voltage is removed when tested as specified in 4.4.4.5(c).

3.3.5.4 Reset. The voltage across pins H and Q shall decrease to less than 1.0 volt when the RESET mode is actuated when tested as specified in 4.4.4.5(d).

3.3.6 Network and amplifier noise level. The noise in the network output and the amplifier output shall not exceed 50 millivolts peak-to-peak when tested as specified in 4.4.4.6.

3.3.7 Trigger level range. The trigger level shall be adjustable from 13.0 to 17.0 millivolts direct current input when tested as specified in 4.3.3.1.

3.3.8 Network time constant. The output voltage of the network shall decrease at least 0.08 volt but no more than 0.33 volt in 1.0 ± 0.1 minute when tested as specified in 4.4.4.7.

3.3.9 Network capacitor leakage. The output of the network shall not change more than 0.05 volt in 2.0 ± 0.2 minute when tested as specified in 4.4.4.8.

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3.3.10 Temperature stability. The following performance shall be attained at +40.0 ± 5.0° F (5°C) and at +140.0 ± 5.0° F (60°C). The electronic module shall be allowed to condition at each of the specified temperatures for at least 15 minutes, after the temperature chamber has stabilized, prior to testing at that temperature for the following requirements.

3.3.10.1 Supply voltage. The output of the 17 - volt supply established in 3.3.3 shall not change more than 200 millivolts when tested as specified in 4.4.4.9.

3.3.10.2 Zero level. The zero shift shall be less than 5.5 millivolts (referred to the input) when tested as specified in 4.4.4.9.

3.3.10.3 Trigger level. The trigger level established in 3.3.5.1 shall not change more than 1.0 millivolt when tested as specified in 4.4.4.9.

3.3.10.4 Network time constant. The network time constant shall meet the requirement of 3.3.8 when tested as specified in 4.4.4.9.

3.4 Operation after vibration. The electronic module shall meet the requirements of 3.3.1 through 3.3.6, 3.3.8, and 3.3.9 after exposure to the vibration environment described in MIL-STD-810, method 514, procedure VIII, curve Y (6,000 miles) when tested as specified in 4.3.3.2. The electronic module shall be visually inspected after vibration and there shall be no physical damage, loose parts, nuts, or bolts.

3.5 Preproduction. Prior to the start of regular production, a preproduction sample of the electronic module assemblies shall be produced in accordance with this specification for examination and test (see 4.3).

3.6 Workmanship. The workmanship of the electronic module assembly shall be in accordance with MIL-STD-252.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 8 electronic module assemblies shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Inspection procedures for examination and tests. The preproduction sample of electronic module assemblies shall be examined in accordance with the classification of defects 4.4.3.3 and tested in accordance with 4.3.3 and 4.4.4.

4.3.3 Tests. Tests shall be conducted as follows:

4.3.3.1 Trigger level range. Determine the trigger level range of the electronic module utilizing the Q199 (DL 136-15-190) in accordance with the instruction manual. The trigger level range shall be as required by 3.3.7.

4.3.3.2 Operation after vibration. Mount the electronic module onto a vibration exciter. No electrical connections need be made; the electronic module need not be operated during the vibration test. Subject the electronic module to the vibration environment required by 3.4. Then, utilizing the Q199 (D136-15-190) in accordance with the instruction manual, test the performance of the electronic module against the requirements of 3.3.1 through 3.3.6, 3.3.8 and 3.3.9. The performance of the electronic module shall be as specified therein. Remove turnlock fastener stud before vibration and mount electronic module assembly through four corner holes. Examine for physical deterioration.

4.3.4 Acceptance/rejection criteria. The preproduction sample of electronic module assemblies shall comply with the examination and tests specified in 4.3.2 to be acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

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4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the electronic module assemblies offered for acceptance at one time which have been produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.4.2 Sampling for examination and for temperature stability tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedures.

4.4.3.1 For examination and for temperature stability tests. Sample electronic module assemblies shall be examined and tested for temperature stability tests in accordance with the classification of defects 4.4.3.3 and MIL-STD-105.

4.4.3.2 For tests except temperature stability. Each electronic module assembly in the lot shall be tested for current drain, battery test circuit, supply voltage and regulation, zero adjust range, alarm functioning, network and amplifier noise level, network time constant, and network capacity leakage in accordance with 4.4.4 to the requirements of 3.3.1 through 3.3.6, 3.3.8 and 3.3.9. Any electronic module found defective shall be rejected.

4.4.3.3 Classification of defects.

(a) Electronic module assembly (Dwg D5-15-4550).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 2.5 percent defective	
101	Temperature stability	4.4.4.9
102	2.225 maximum width of wiring board	
103	.184 ± .005 distance from edge of wiring board to centerline of guide pin of electrical connector	
104	.925 ± .015 distance from centerline of guide pin of electrical connector to centerline of turnlock fastener shaft	
105	3.207 - .014 distance from face of electrical connector to cover plate	

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4.4.3.3 Classification of defects (continued).

(a) Electronic module assembly (Dwg D5-15-4550).

Categories	Defects	Acceptance standards
Major:		
106	3.417 + .009 distance from centerline of pin on turnlock fastener shaft to cover plate	
107	2.06 maximum width of case assembly	
108	.125 ± .009 distance from centerline of electrical connector guide pins to wiring board	
109	.070 maximum thickness of wiring board at mating edges	
110	.468 ± .014 distance from centerline of turnlock fastener shaft to centerline of electrical connector guide pins	
111	.984 maximum height of case assembly from wiring board	
112	14° ± 5° angular relationship between centerline of pin and centerline of turnlock fastener shaft	
113	Workmanship (applicable defects listed as Major in MIL-STD-252, see 3.6)	
Minor:	AQL 4.0 percent defective	
201	Workmanship (applicable defects listed as Minor in MIL-STD-252, see 3.6)	
	AQL 6.5 percent defective	
202	Workmanship (applicable defects listed as Control in MIL-STD-252, see 3.6)	

(b) Printed wiring board (Dwg D5-15-5568).

Categories	Defects
Critical:	None defined

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4.4.3.3 Classification of defects (continued).

(b) Printed wiring board (Dwg D5-15-5568).

Categories	Defects
Major: AQL 2.5 percent defective	
101	.055 maximum height of projection on underside of wiring board

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Current drain. Determine the current drain of the electronic module utilizing the Q199 (D136-15-190) in accordance with the instruction manual. The current drain shall be as required by 3.3.1.

4.4.4.2 Battery test circuit. Determine the current through the battery test circuit of the electronic module utilizing the Q199 in accordance with the instruction manual. The current through the battery test circuit shall be as required by 3.3.2.

4.4.4.3 Supply voltage and regulation. Determine the output supply voltage and regulation of the electronic module utilizing the Q199 in accordance with the instruction manual. The output shall be as required by 3.3.3.

4.4.4.4 Zero adjust range. Determine the zero adjust range of the electronic module utilizing the Q199 in accordance with the instruction manual. The zero adjust range shall be as required by 3.3.4.

4.4.4.5 Alarm functioning.

(a) Actuation. Determine the alarm actuation of the electronic module utilizing the Q199 in accordance with the instruction manual. The alarm actuation shall be as required by 3.3.5.1.

(b) Alarm level stability. Determine the alarm level stability of the electronic module utilizing the Q199 in accordance with the instruction manual. The alarm level stability shall be as required by 3.3.5.2.

(c) Latching. Determine the alarm latching ability of the electronic module utilizing the Q199 in accordance with the instruction manual. The alarm latching ability shall be as required by 3.3.5.3.

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(d) Reset. Determine the operation of the reset mode of the electronic module utilizing the Q199 in accordance with the instruction manual. The reset mode shall operate as required by 3.3.5.4.

4.4.4.6 Network and amplifier noise level. Determine the network and amplifier noise level of the electronic module utilizing the Q199 in accordance with the instruction manual. The network and amplifier noise level shall be as required by 3.3.6.

4.4.4.7 Network time constant. Determine the network time constant of the electronic module utilizing the Q199 in accordance with the instruction manual. The network time constant shall be as required by 3.3.8.

4.4.4.8 Network capacitor leakage. Determine the network capacitor leakage of the electronic module utilizing the Q199 in accordance with the instruction manual. The network capacitor leakage shall be as required by 3.3.9.

4.4.4.9 Temperature stability. Determine the temperature stability of the electronic module utilizing the Q199 in accordance with the instruction manual. The temperature stability shall be as required by 3.3.10.

5. PACKAGING

5.1 Packaging, packing and marking, interplant shipment (see 6.3). The electronic module shall be packaged and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with rules and regulations applicable to the mode of transportation. Marking shall be in conformance with Fed-Std-123 and shall include the contractors name, address and contract number.

5.2 Repair parts. When this item is being procured for storage and issue as a repair part, preservation, packaging, packing and marking shall be as specified on the repair parts packaging data sheet which is identified by its National stock number.

6. NOTES

6.1 Intended use. This electronic module assembly is used to complete a function as a subassembly in the detector unit which is used in a series of chemical agent alarms.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

(1) Time allowed for contractor submission of sample for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Interplant shipment. Packaging and packing for interplant shipment is for supplies and material that do not enter the military supply system. Typical interplant shipments are shipments from a vendor to a contractor, or from a contractor to a military arsenal or plant.

6.4 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 251. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

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